

AI Doctor

An AI-Powered Intelligent Health Symptom & Wellness Mentor

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THE CHALLENGE

Problem Statement

People experiencing everyday symptoms often struggle to find quick, reliable guidance — leading to anxiety, delayed care, or unnecessary hospital visits.



No Quick, Trustworthy Guidance

Generic search results are overwhelming, inconsistent, or medically unreliable.



Difficulty Understanding Symptoms

Users struggle to connect multiple symptoms to likely, common causes.



Lack of Structured Self-Care Advice

Little clarity on what to do at home versus when professional care is needed.



Delayed Awareness of Warning Signs

Users may miss red flags that indicate the need for urgent medical attention.



No Personalized Health Follow-up

No easy way to track symptoms over time or get tailored follow-up guidance.



Limited Access to Instant Support

Doctor consultations aren't always immediately available for quick questions.

"People need an intelligent companion instead of just another search engine."

What AI Doctor Aims to Achieve

A comprehensive set of objectives that transform how people understand symptoms and approach everyday health decisions.

Understand User Symptoms

Interpret free-text symptom descriptions clearly and contextually.

Assess Possible Causes

Identify likely, common causes behind reported symptom combinations.

Generate Self-Care Guidance

Offer practical, safe home-care steps for mild, everyday conditions.

Flag Warning Signs

Highlight symptoms that may need urgent professional attention.

Explain Conditions Clearly

Break down possible conditions with simple, easy-to-understand language.

Provide Wellness Tips

Offer hydration, rest, nutrition, and recovery guidance.

Track Symptom History

Help users log and monitor symptoms across multiple check-ins.

Recommend Next Steps

Suggest whether to rest, monitor, or consult a doctor.

Encourage Professional Care

Always reinforce timely consultation with certified medical professionals.

SOLUTION OVERVIEW

Introducing AI Doctor

AI Doctor acts as an intelligent health companion — not a diagnosis tool — capable of understanding your symptoms, offering clear guidance, and helping you decide the right next step.

AI Doctor combines IBM Watson Orchestrate automation with advanced LLM reasoning to deliver clear, safe, and reassuring health guidance for every user.

This tool provides general health information only and does not replace professional medical diagnosis or treatment.

How the AI Doctor Agent Works

A seamless, end-to-end intelligent workflow that takes a user from symptom entry all the way to clear guidance.

Profile Setup

User registers and shares basic health background.

Symptom Intake

AI gathers current symptoms through guided, adaptive questions.

Severity Check

User indicates duration, intensity, and any red-flag signs.

Possible Causes

AI generates a structured list of likely, common causes.

Self-Care Guidance

Practical home-care steps tailored to reported symptoms.

Condition Explanation

Clear explanations of related conditions in plain language.

Risk Flagging

AI highlights symptoms that need urgent medical attention.

Doctor Recommendation

Suggests consultation urgency and specialist type if needed.

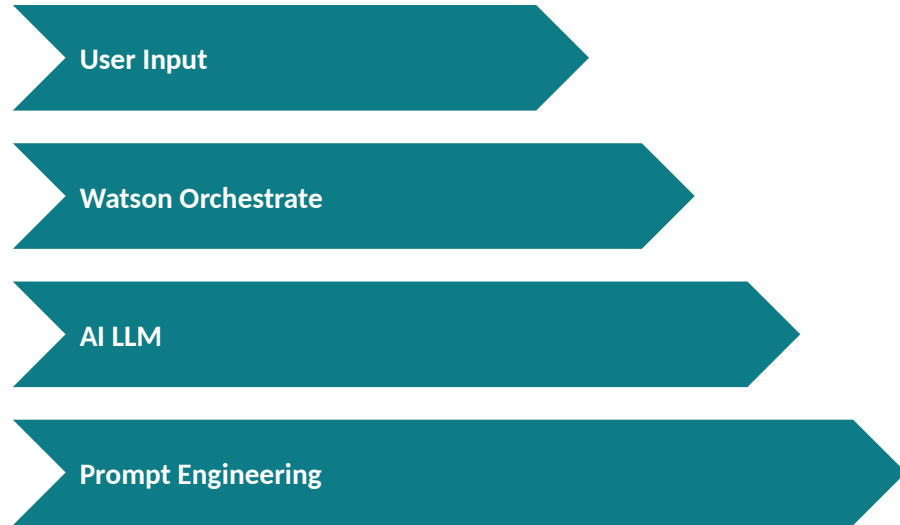


Progress Tracking — Continuous monitoring of symptoms, patterns, and follow-up reminders.



Care Readiness — Clear next-step recommendation and doctor referral when needed.

How AI Doctor Works



Intelligence at Every Layer

Understands User Symptoms

IBM Watson Orchestrate parses symptom details, duration, and context from every interaction.

Generates Intelligent Guidance

The LLM layer produces accurate, context-rich explanations of possible causes and care steps.

Uses Contextual Prompt Engineering

Carefully crafted prompts ensure responses are medically cautious and appropriately worded.

Delivers Personalized Recommendations

The decision engine tailors guidance and urgency level to each user's symptoms.

Continuously Adapts to Follow-ups

The system learns from every check-in to refine guidance and track progress.

Key Features That Power AI Doctor



Symptom Checker

Understands multi-symptom descriptions and identifies likely common causes.



Condition Explainer

Clear, step-by-step explanations of common conditions in plain language.



Self-Care Recommender

Practical home-care suggestions based on symptom type and severity.



Smart Follow-up Questions

Adaptive questions that narrow down possible causes accurately.



Urgency Analyzer

Flags red-flag symptoms and estimates how soon to seek care.



Doctor Referral Guide

Suggests the right specialist type based on reported symptoms.



Symptom History Log

Tracks reported symptoms and patterns across multiple check-ins.



Wellness Coach

Offers hydration, nutrition, sleep, and recovery guidance.



Medication Reminder

Personalized reminders to support recovery and daily wellness.



AI Doctor 

Model: GPT-OS

Talk to agent

 Running in debug mode

10:29 PM

Hello, welcome to AI Doctor, your intelligent health companion...

Accuracy of generated answers may vary. Please double-check responses.

What causes headaches?

Check my symptoms

General wellness tips

Profile

 Notifications: 1 Dismissed | 1 total

Profile

Describe what your agent does so people and other agents know when to use it.

Behavior

AI Doctor is an AI-powered health assistant that helps users understand possible causes of common symptoms, offers general self-care and wellness guidance, tracks symptom check-ins, and encourages timely consultation with certified medical professionals.

Welcome message

Customize the welcome message from the agent on the home screen.

Type something...



AI Doctor 

Profile

Knowledge

 Behavior

Define how and where your agent should react to requests and respond to users.

Behavior
Instructions

Help users understand possible causes of their symptoms, provide general self-care guidance, and always recommend consulting a certified doctor for diagnosis or treatment.

Guidelines

Add Guideline +

Model: GPT-OS

Talk to agent

 Running in debug mode

Today

What causes nausea, headache, and mild fever?

AI Doctor 10:31 PM

Possible Causes of Nausea, Headache & Mild Fever

These three symptoms together are commonly linked to:

- Viral infection / flu**
Most common cause of this combination.
- Migraine**
Can cause nausea, light sensitivity, and low-grade fever.
- Food poisoning or indigestion**
Often comes with stomach discomfort.
- Dehydration**
Reduced fluid intake can trigger all three symptoms.

 Stay hydrated and monitor your temperature closely.



AI Doctor 

Profile

Knowledge

↑ Behavior

Instructions
Behavior

Help users build clear understanding of their symptoms, provide safe self-care steps, flag warning signs, and always encourage timely consultation with a certified doctor for diagnosis or treatment.

Model: GPT-OS

Talk to agent

 Running in debug mode

You 10:32 PM

Symptoms: Nausea, headache, mild fever
Duration: 1 day
Severity: Mild
Ask: What causes this and what should I do?

AI Doctor 10:32 PM

Here's What to Do Next

Home Care

Rest, sip water or ORS often, eat light meals, and monitor temperature every few hours.

See a Doctor If

Fever rises above 102°F, symptoms last more than 2–3 days, or severe dehydration develops.



This is general information only, not a medical diagnosis. Please consult a healthcare professional.

AI Doctor — Use Cases at a Glance

Every feature of AI Doctor is designed to solve a specific user pain point and deliver measurable health-guidance outcomes.

Use Case	Functionality	User Benefit	Outcome
Symptom Checker	Identifies likely causes from symptom combinations	Faster clarity on health concerns	Reduced anxiety
Condition Explainer	Explains conditions with plain-language detail	Better health understanding	Informed decisions
Self-Care Recommender	Suggests safe home-care steps	Practical, actionable guidance	Faster recovery
Urgency Analyzer	Flags red-flag / warning symptoms	Timely awareness of risk	Safer outcomes
Doctor Referral Guide	Recommends the right specialist type	Efficient care pathway	Better placement of care
Symptom History Log	Tracks symptoms across check-ins	Visibility into patterns	Proactive health tracking
Wellness Coach	Hydration, rest & nutrition guidance	Improved daily wellness	Healthier habits
Follow-up Reminders	Personalized recovery check-ins	Consistent monitoring	Better adherence

Health Impact & Real-World Applications

Health Impact

→ Personalized Health Guidance

Every user receives symptom guidance tailored to their unique profile.

→ Faster Symptom Clarity

Intelligent explanations eliminate confusion and reduce unnecessary worry.

→ Improved Self-Care Confidence

Clear home-care steps build confidence in everyday health decisions.

→ Earlier Risk Awareness

Warning-sign flagging encourages timely medical consultation.

→ Better Health Outcomes

Structured guidance directly supports faster, safer recovery.

Real-World Applications

- Personal daily health checks
- Telehealth pre-consultation triage
- Corporate wellness programs
- Rural & remote health access
- Elderly care support
- Campus health centers

Improved early awareness and better-informed healthcare decisions across communities.

AI Doctor — The Future of Everyday Health Guidance

AI Doctor transforms everyday symptom confusion into clear, personalized, and safe guidance by combining AI-powered reasoning with workflow automation through IBM Watson Orchestrate.

Future Enhancements

01

Wearable Integration

Sync with fitness trackers for real-time vitals context.

02

Multilingual Support

Offer guidance in multiple regional languages.

03

Voice-based Interaction

Enable hands-free, conversational symptom checks.

04

Medication Interaction Checker. Live Healthier with AI Doctor."

Flag potential interactions between medicines.

Coming Next

01

Telehealth Handoff

Direct connection to licensed doctors for urgent cases.

02

Chronic Condition Tracking

Long-term monitoring for ongoing health conditions.

03

Family Health Profiles

Manage symptom history for multiple family members.

04

AI Health Dashboard. Anasneem • J.N.N Institute of Engineering • B.E CSE • 2026-2027

Visual analytics on symptom trends and wellness scores.